



COURSE SLIDES

Certified Lean Six Sigma White Belt (CLSSWB) Training

WSQ Course Code: TGS-2025053210

Conducted by Tertiary Infotech Academy Pte Ltd | UEN 201200696W

Version 1

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About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on AI agents, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Agenda

- Day 1: Lean Six Sigma foundations, customer value, process walk, waste, basic data
- Day 2: 5 Whys, Fishbone, 5S, standard work, visual controls, improvement summary
- Final sharing, assessment readiness, feedback and closing

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

Course Objectives

- Explain Lean, Six Sigma, DMAIC, and the White Belt support role.
- Recognize value, waste, handoffs, waiting, defects, and variation.
- Use simple process maps, check sheets, charts, 5 Whys, Fishbone, 5S, and action plans.
- Prepare a small improvement portfolio based on the lab sequence.

Digital Attendance and LMS

- Take AM, PM, and assessment digital attendance when instructed.
- Access courseware, assessment information, and submissions on the LMS.
- Minimum attendance requirements apply for WSQ funding and assessment eligibility.
- Inform the trainer immediately if you cannot access the LMS or attendance QR.

Assessment Overview

- Assessment checks understanding of White Belt concepts and practical lab evidence.
- Keep all lab deliverables in your improvement portfolio.
- Use the Learner Guide and lab outputs to prepare for final review.
- Competency depends on completing required assessment instructions.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Portfolio Rules

- Use one simple process throughout the course where possible.
- Save each lab output using clear names.
- Capture evidence from observation, simple data, and peer/trainer feedback.
- Escalate complex or sensitive issues to a process owner, manager, Green Belt, or Black Belt.

SECTION

Lean Six Sigma Foundations

Shared language for value, waste, variation, and DMAIC

Lean and Six Sigma

Lean

- Increase customer value
- Reduce waste and improve flow
- Make work easier to see and perform

Six Sigma

- Reduce defects and variation
- Use data and structured problem solving
- Improve process capability over time

DMAIC: The White Belt Roadmap



White Belts do not need to lead DMAIC projects; they help the team use shared language, observe work, and keep evidence organized.

White Belt Role

- Participates in improvement conversations and lab activities.
- Observes work, records simple facts, and helps collect basic data.
- Identifies waste and improvement opportunities.
- Supports implementation of agreed standard work and visual controls.
- Escalates issues outside the White Belt scope.

Lab 1: Lean Six Sigma Foundations

- Explain Lean and Six Sigma in plain language.
- Describe DMAIC.
- Understand the White Belt support role.
- Select a simple process for the course.

Lab 1: Scenario

- You are new to Lean Six Sigma and have been asked to support improvement activity in your team. Your first task is to understand the basic language and choose a simple process to observe.

Lab 1: Step-by-Step (1/2)

- Write a one-sentence definition of Lean.
- Write a one-sentence definition of Six Sigma.
- List the five DMAIC phases.
- Write one simple example activity for each DMAIC phase.
- List three ways a White Belt can support an improvement project.

Lab 1: Step-by-Step (2/2)

- List two situations that should be escalated to a Green Belt, Black Belt, manager, or process owner.
- Choose one simple workplace process.
- Write why the process matters to customers, learners, staff, or the business.
- Write one problem you have seen in the process.
- Share your process example with another learner.

Lab 1: Validation and Evidence

Validation checks

- Your Lean and Six Sigma definitions are understandable.
- You can explain DMAIC without reading from notes.
- Your selected process is simple enough to observe.
- You know the limits of the White Belt role.

Deliverables

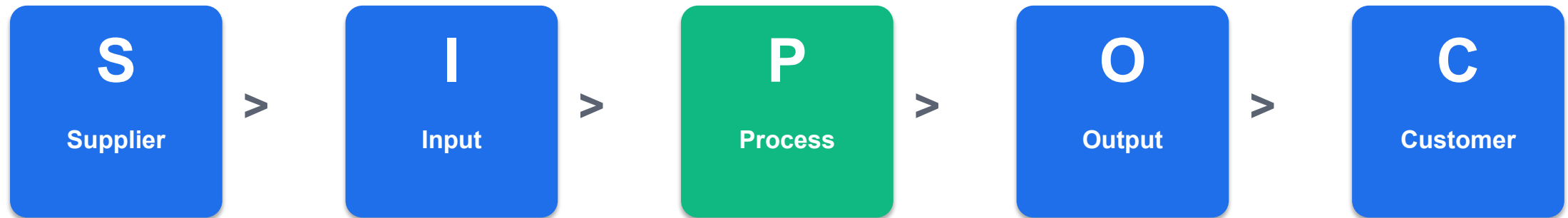
- Lean and Six Sigma definitions.
- DMAIC summary.
- White Belt role notes.
- Selected process example.

SECTION

Customer Value and Process Thinking

Define customers, outputs, boundaries, and flow

SIPOC: Define the Process Boundary



- Start with the customer and the process output.
- Keep the map simple enough for a short discussion.
- Use SIPOC before detailed process mapping.

Lab 2: Customer Value and Process Thinking

- Identify process customers.
- Define customer value.
- Build a simple SIPOC.
- Identify value-added and non-value-added activity.

Lab 2: Scenario

- A process exists to deliver value to someone. You will identify the customer, what they need, and the process boundary before looking for waste.

Lab 2: Step-by-Step (1/3)

- Write the name of your selected process.
- Identify the customer or receiver of the process output.
- Write what the customer needs from the process.
- Define the process start point.
- Define the process end point.

Lab 2: Step-by-Step (2/3)

- List suppliers.
- List inputs.
- List 5 to 7 major process steps.
- List outputs.
- List customers.

Lab 2: Step-by-Step (3/3)

- Create a simple SIPOC table.
- Mark each major step as value-added, business-required, or non-value-added.
- Circle one step that looks slow, unclear, repeated, or error-prone.

Lab 2: Validation and Evidence

Validation checks

- The customer is clearly identified.
- SIPOC has a clear start and end boundary.
- Activity labels are based on customer value.
- One improvement opportunity is identified.

Deliverables

- Customer value worksheet.
- Simple SIPOC.
- Value-added activity list.
- Improvement opportunity note.

SECTION

Process Observation and Waste

Observe work respectfully and identify TIMWOODS waste

TIMWOODS Waste

- Transport: unnecessary movement of items or information
- Inventory: excess work-in-progress or backlog
- Motion: unnecessary movement by people
- Waiting: idle time or delayed handoffs
- Overproduction: work done before it is needed
- Overprocessing: extra checking or unnecessary steps
- Defects: errors, rework, missing or incorrect information
- Skills: underused ideas, knowledge, or capability

Lab 3: Process Walk and Simple Map

- Observe a process respectfully.
- Record actual work steps.
- Draw a simple process map.
- Identify handoffs and waiting.

Lab 3: Scenario

- Processes often work differently from how people describe them. You will observe the real workflow and map what actually happens.

Lab 3: Step-by-Step (1/3)

- Review the process from Lab 02.
- Visit the place where work happens or interview someone who performs the work.
- Observe without blaming, interrupting, or correcting people.
- Record the actual steps in order.
- Record who performs each step.

Lab 3: Step-by-Step (2/3)

- Note handoffs between people, teams, tools, or systems.
- Note waiting points.
- Note rework, repeated checking, or unclear decisions.
- Draw a simple flowchart using boxes and arrows.
- Add decision diamonds where the process branches.

Lab 3: Step-by-Step (3/3)

- Mark one process step that should be discussed with the process owner.
- Ask another learner whether your map is easy to follow.

Lab 3: Validation and Evidence

Validation checks

- The map follows the actual process sequence.
- Handoffs and waits are visible.
- Observations are factual and respectful.
- The map can be understood by someone outside the process.

Deliverables

- Process walk notes.
- Simple process map.
- Handoff and waiting notes.
- Process owner discussion point.

Lab 4: Identify TIMWOODS Waste

- Recognize the eight wastes.
- Find waste in a simple process.
- Link waste to customer impact.
- Prioritize one waste area.

Lab 4: Scenario

- Lean improvement starts with seeing waste. You will review your process map and identify where time, effort, information, or talent is being wasted.

Lab 4: Step-by-Step (1/4)

- Review TIMWOODS:
- Transport.
- Inventory.
- Motion.
- Waiting.

Lab 4: Step-by-Step (2/4)

- Overproduction.
- Overprocessing.
- Defects.
- Skills.
- Open the process map from Lab 03.

Lab 4: Step-by-Step (3/4)

- Review each process step one by one.
- Mark examples of unnecessary movement of items or information.
- Mark queues, backlog, or work-in-progress.
- Mark unnecessary movement by people.
- Mark waiting time.

Lab 4: Step-by-Step (4/4)

- Mark work done too early or in too much quantity.
- Mark extra checking or repeated entry.
- Mark errors, rework, missing information, or corrections.
- Mark places where people's ideas or skills are not used.
- Choose the one waste area with the largest visible impact.

Lab 4: Validation and Evidence

Validation checks

- Waste examples are tied to specific process steps.
- The priority waste has a clear customer or business impact.
- Observations are specific enough for discussion.
- The team avoids blaming individuals.

Deliverables

- TIMWOODS waste table.
- Marked process map.
- Priority waste statement.
- One improvement question.

SECTION

Basic Data and Visuals

Use simple counts and charts to support better discussions

Check Sheet to Chart

- Choose one issue that can be counted safely.
- Use clear categories so everyone counts the same thing.
- Tally occurrences consistently.
- Sort categories from largest to smallest.
- Use a bar chart, Pareto-style view, or run chart to start the conversation.

Lab 5: Basic Data and Visuals

- Create a basic check sheet.
- Count issue categories.
- Build a simple visual.
- Use data to focus discussion.

Lab 5: Scenario

- Improvement should not rely only on opinion. You will collect a small amount of simple data and use it to identify what to discuss first.

Lab 5: Step-by-Step (1/3)

- Choose one issue or waste area from Lab 04.
- Decide what can be counted safely.
- Define clear categories.
- Create a check sheet with category names and tally marks.
- Collect sample data or use a trainer-provided dataset.

Lab 5: Step-by-Step (2/3)

- Add the tally marks.
- Calculate category totals.
- Calculate total observations.
- Create a bar chart.
- Sort the chart from largest category to smallest category.

Lab 5: Step-by-Step (3/3)

- Create a run chart if the data includes dates or times.
- Write one sentence explaining what the visual shows.
- Write one question that should be investigated.

Lab 5: Validation and Evidence

Validation checks

- Categories are clear and not overlapping.
- Counts are added correctly.
- The chart has a title and labels.
- The conclusion is based on the data shown.

Deliverables

- Check sheet.
- Tally table.
- Bar chart or Pareto-style chart.
- Run chart if time data is available.
- Data observation note.

SECTION

Cause Discussion and Improvement Control

Move from possible causes to practical next steps

5 Whys and Fishbone

5 Whys

- Start with an observable problem statement.
- Ask why the condition happens.
- Avoid blame and stop when evidence is missing.

Fishbone

- Group possible causes by category.
- Use people, process, tools, materials, measurement, and environment.
- Circle causes needing confirmation.

Lab 6: Problem Solving with 5 Whys and Fishbone

- Write a fact-based problem statement.
- Use 5 Whys.
- Build a Fishbone diagram.
- Identify evidence needed before action.

Lab 6: Scenario

- The team has identified a visible issue. Before suggesting solutions, you will explore possible causes in a structured and respectful way.

Lab 6: Step-by-Step (1/3)

- Select the issue from Lab 05.
- Write a problem statement using facts from the process or data.
- Ask why the problem happens.
- Continue asking why until you reach a process-related possible cause.
- Avoid blaming a person as the final cause.

Lab 6: Step-by-Step (2/3)

- Create a Fishbone diagram.
- Use categories such as people, process, tools, materials, measurement, and environment.
- Add possible causes to each category.
- Circle two causes that seem most likely.
- Write what evidence would support or reject each cause.

Lab 6: Step-by-Step (3/3)

- Identify who can help confirm the cause.
- Write one safe next investigation step.

Lab 6: Validation and Evidence

Validation checks

- The problem statement is factual.
- The 5 Whys chain does not stop at blame.
- Fishbone causes are plausible.
- Evidence needed is clearly listed.

Deliverables

- Problem statement.
- 5 Whys worksheet.
- Fishbone diagram.
- Evidence-needed list.
- Next investigation step.

5S, Standard Work, and Visual Controls

- Sort: remove what is not needed.
- Set in Order: make needed items easy to find and use.
- Shine: clean, correct, and clarify.
- Standardize: define the improved way of working.
- Sustain: check that the standard continues.
- Use labels, checklists, templates, status boards, and simple visual cues.

Lab 7: 5S, Standard Work, and Visual Controls

- Apply 5S thinking.
- Identify visual control opportunities.
- Draft simple standard work.
- Support sustainment.

Lab 7: Scenario

- Small improvements are easier to sustain when the workplace is organized, the standard is clear, and the team can see when something is not normal.

Lab 7: Step-by-Step (1/3)

- Choose one physical workspace, digital folder, form, queue, or process step.
- Apply Sort: list what is needed and not needed.
- Apply Set in Order: decide where needed items or information should be located.
- Apply Shine: identify what should be cleaned, corrected, clarified, or updated.
- Apply Standardize: define what good looks like.

Lab 7: Step-by-Step (2/3)

- Apply Sustain: define how the standard will be checked.
- Identify where a label, checklist, color code, status board, or template would help.
- Draft a short standard work instruction.
- Include task name, steps, expected result, and check method.
- Ask a partner to review whether the instruction is clear.

Lab 7: Step-by-Step (3/3)

- Revise the standard work based on feedback.

Lab 7: Validation and Evidence

Validation checks

- The 5S actions are practical.
- The visual control makes abnormal conditions easier to see.
- Standard work is written simply.
- Another learner can follow the instruction.

Deliverables

- 5S action list.
- Visual control idea.
- Draft standard work.
- Feedback notes.

Lab 8: Improvement Summary and Action Plan

- Summarize the improvement opportunity.
- Create a simple action plan.
- Identify who should be involved next.
- Present a White Belt improvement story.

Lab 8: Scenario

- You have observed a process, identified waste, collected simple data, and explored possible causes. You will now summarize the opportunity and prepare a practical next step.

Lab 8: Step-by-Step (1/3)

- Review all lab worksheets.
- Write the process name.
- Identify the customer.
- Summarize the problem in one sentence.
- Summarize the waste observed.

Lab 8: Step-by-Step (2/3)

- Summarize the simple data collected.
- Summarize the possible cause.
- Write one practical improvement idea.
- Identify the process owner or team that should review the idea.
- Define the first action.

Lab 8: Step-by-Step (3/3)

- Assign an owner or recommended owner.
- Set a due date or review date.
- Define how the team will know whether the change helped.
- Prepare a three-minute presentation.
- Present to a peer or trainer and record feedback.

Lab 8: Validation and Evidence

Validation checks

- The summary uses evidence from previous labs.
- The action plan has owner, date, and measure.
- The proposed improvement is small and realistic.
- The presentation is clear and respectful.

Deliverables

- Improvement summary.
- Action plan.
- Measurement idea.
- Presentation notes.
- Lessons learned.

Final Portfolio Checklist

- Lean/Six Sigma definitions and DMAIC notes
- Customer value worksheet and simple SIPOC
- Process map and TIMWOODS waste table
- Check sheet, simple chart, and data observation
- Problem statement, 5 Whys, Fishbone, and evidence-needed list
- 5S action list, standard work draft, improvement summary, and action plan

Take the Online CLSSWB Practice Questions

- Reinforce Lean, Six Sigma, DMAIC, White Belt role and improvement basics before assessment day.
- Use practice sets to test process thinking, TIMWOODS waste, data visuals, 5 Whys, Fishbone and 5S.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to the review session.

Access and practise at:

Google Classroom > Practice Questions

[Tertiary Exams](#)[Practice Exam](#)[About Us](#)[Get started](#)

Certified Lean Six Sigma White Belt Practice Questions

MODE
Practice

FOCUS
CLSSWB

ACCESS
Class link

PRACTICE DETAILS
Exam code WB-001
Use Review
Domains 5

Practice mode
Check answers and explanations while revising each topic.

DMAIC and White Belt role		34%
Waste and process thinking		28%
Data and problem solving		23%
5S and action planning		15%

Learner tip
Retake missed items after revision.

THANK YOU

Closing

- Complete feedback or TRAQOM survey as instructed.
- Confirm final attendance and assessment submission requirements.
- Apply one small improvement conversation back at work.